

Week 6

Keel Bone Chicken Project

06/29/26



College of Engineering

This Week

Weekly Progress

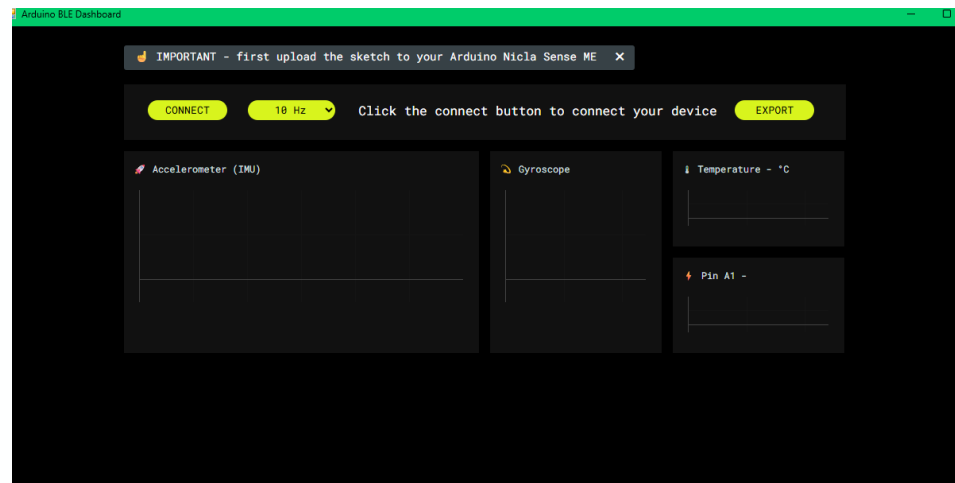
- Database Updates
 - Local Hosting Enabled
 - Expanded Test Code from Last Week
- Web Logger Updates
 - MongoDB UI now in use
 - Data Export Functional Fixes
- Supply Updates
 - Shipment dates
- Testing Updates
 - Characterization Sent
- Next Week

Database Updates

Local Hosting Enabled

- MongoDB Key is now optional
- Can be hosted on local machine (without use of web application)
- Need to walk Agriculture department through the steps
- Example of Mongo URI Key below:

```
Web App > NiclaSenseME-dashboard > .env.example
1 MONGODB_URI=mongodb://huangnathan69_db_user:BigFuture200@ac-smzxmrw-shard-00-00.pclc27w.mongodb.net:27017,ac-smzxmrw-shard-0
2
```



Running as GUI
application now

Database Updates

Expanded Test Code from Last Week

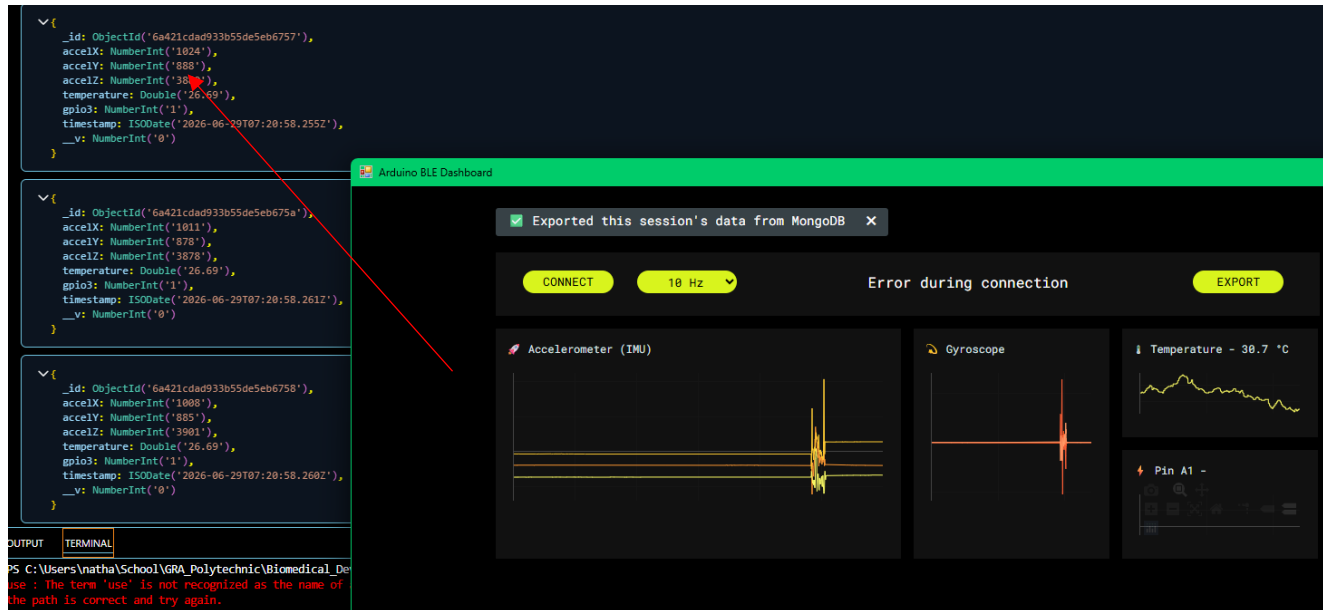
- Expanded Simple Test Code from Last Week
- Got working version of test code → working logger on localhost
- Code seen below:

```
160     w = quaternion.w();
161
162     float quaternionValues[] = {x,y,z,w};
163     quaternionCharacteristic.writeValue(quaternionValues, sizeof(quaternionValues));
164 }
165
166 }
167
168
169 void blePeripheralDisconnectHandler(BLEDevice central){
170     nicla::leds.setColor(red);
171 }
172
173 void onTemperatureCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
174     float temperatureValue = temperature.value();
175     temperatureCharacteristic.writeValue(temperatureValue);
176 }
177
178 void onHumidityCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
179     uint8_t humidityValue = humidity.value() + 0.5f; //since we are truncating the float type to a uint8_t, we want to round it
180     humidityCharacteristic.writeValue(humidityValue);
181 }
182
183 void onPressureCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
184     float pressureValue = pressure.value();
185     pressureCharacteristic.writeValue(pressureValue);
186 }
187
188 void onBsecCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
189     float airQuality = float(bsec.iaq());
190     bsecCharacteristic.writeValue(airQuality);
191 }
192
193 void onCo2CharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
194     uint32_t co2 = bsec.co2_eq();
195     co2Characteristic.writeValue(co2);
196 }
```

Web Logger Updates

MongoDB UI now in use

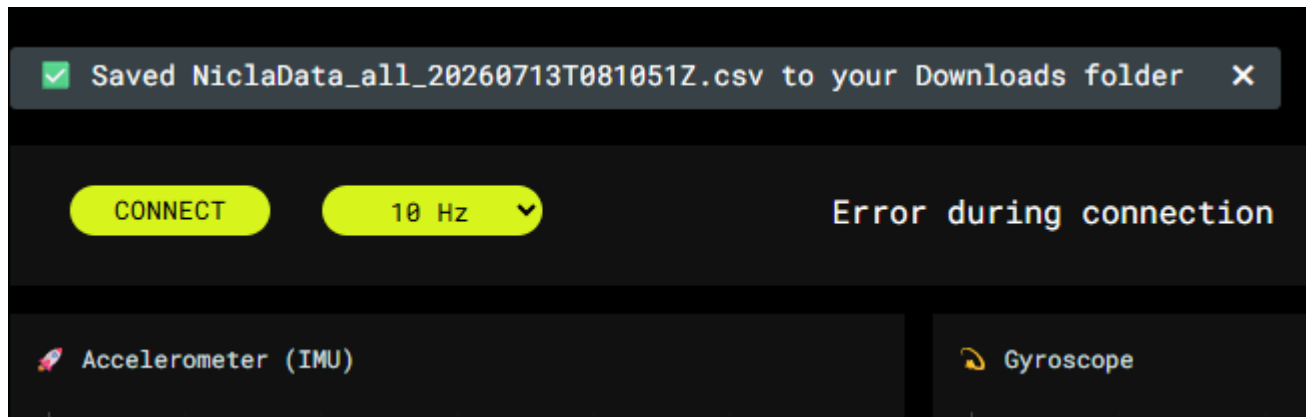
- MongoDB UI in use (over MongoDB Atlas)
- Connection string is now: localhost:12701
- Image of data being loaded seen below



Web Logger Updates

Data Export Functional Fixes

- Fixed functional issue where download would go to unexplored path
- Fixed logger download time to sync up with real world time



Name	Date modified	Type	Size
Today			
 NiclaData_all_20260713T081015Z.csv	7/13/2026 4:10 AM	Microsoft Excel Com...	1 KB

Supply Updates

Shipment Dates

- First purchase request filled (PO 4501590095 UPS)
 - Shipment should have arrived last Friday, need to verify with Professor Karcher
- Second pending (Verified that the number is correct)
 - This shipment should arrive by the end of the week given it goes through approvals quickly

Feed (5)

☒ Edits ☒ Status Changes ☒ Comments



Nathan Huang

Changed Status from "Awaiting Additional Information" to "In Process".

Yes, that number is correct. Feel free to order additional ones if that may be beneficial, but that number is correct.

-Nathan

Mon 7/13/2026 2:59 AM

[Comment](#) [Like](#)



Nathan Bryant

Hello Nathan Huang,

Just following up:

How many of these items are you requesting?

I wanted to make sure that the 20 that are listed in your ticket are correct?

Thank you,

Feed (2)

[Comment](#)

☒ Edits ☒ Status Changes ☒ Comments



Nathan Bryant

Changed Status from "New" to "Closed".

Hello,

Your Ariba order has been completed and is now with the vendor for fulfillment. Once these items are received, please scan and email the packing slips to engrpslip@purdue.edu or submit [here](#) within 24 hours.

If you have any questions or concerns regarding this order, please reply to this ticket and reference your PO [4501590094].

If you have questions on the estimated shipping date, you may contact the vendor directly. Please do not change anything on the order if you call/email for shipping information.

Thank you,
Engineering/Polytechnic Purchasing Center

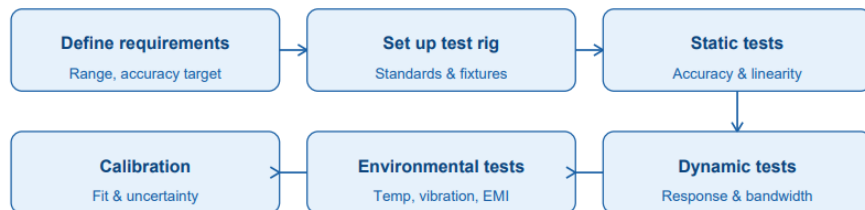
Testing Updates

Characterization Sent

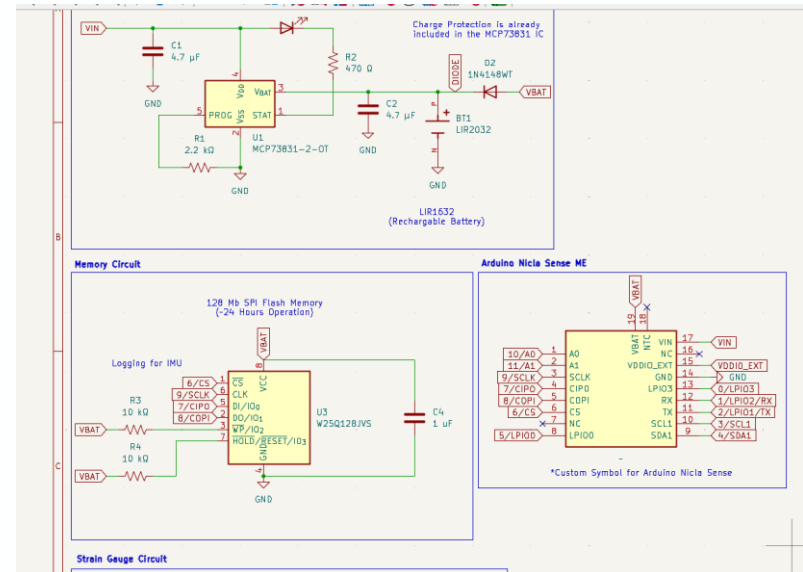
- Pedro requested characterization of the Sensors
- Sent as rudimentary diagram (see below)
- Also sent as PNG for easier system design view
- After approval, should plan for Friday to test sensors

Sensor characterization methodology

Force sensor | Strain gauge | Temperature sensor | IMU



General six-stage methodology applied to all sensor types



Next Week

Goals for next week

- Check on shipment and hardware check with Professor Karcher/Omosalewa
- Verify characterization method with Pedro
- Continue work on database with application efforts

Thank You

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